Do edits that fool one face recognition model also fool models they were never built for?

COMS4036A Project Proposal

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Problem

Tools such as Fawkes, reported at over 500,000 downloads, and LowKey protect people from face recognition by making small edits to photographs before they are posted. Both rely on one assumption: an edit built against one recognition system will work against whatever system a stranger is running. Radiya-Dixit and Tramèr (ICLR 2022) challenge this, since a photograph is edited once and must then defeat every system that follows. Those tools corrupt a model during training; we ask a simpler question.

Our question. We edit a photograph until one model stops recognising it as the same person. Do models we had no access to while making that edit also stop recognising it? Where some resist and others give way, is that explained by their model architecture or by the faces they were trained on?

Data

Labeled Faces in the Wild (LFW) holds 13,233 photographs of 5,749 people. Its standard test, View 2, gives 6,000 fixed pairs, half showing one person twice and half showing two people, scored by ten-fold cross validation. We use these published pairs rather than devising our own split.

LFW is too thin to train on. Only 1,680 of its people appear more than once, leaving 4,069 with a single photograph. The DeepID2 authors state that LFW works as a test set but is inadequate for training, so they train on a larger separate dataset, as will we. The maintainers list six View 2 pairs carrying the wrong label and ask that they be left uncorrected, which we will do.

LFW is built from news photographs of well known people and its make-up is uneven: Han and Jain (2014) estimated it at 77.5% male and 83.5% White using an automatic classifier. Since this project concerns privacy we discuss that in the report, and if time allows we check whether protection is even across groups.

Method

We train a face recognition network from scratch, which learns to place photographs of one person close together and different people far apart. It is the only model used while building edits. Four further models are used for testing only, each differing in one way.

1. InceptionResnetV1 trained on VGGFace2, from `facenet-pytorch`.
2. InceptionResnetV1 trained on CASIA-WebFace. Same design, different faces.
3. ArcFace from InsightFace. Different architecture, different training objective.
4. Eigenfaces or LBPH from Chapter 1, using no deep learning.

We record every model's accuracy on untouched pairs, then edit one photograph in each same-person pair using projected gradient descent, which repeatedly nudges pixels in the direction that makes our model see the photographs as more different, pulling any pixel that drifts too far back inside a fixed limit. That limit is the edit budget, swept from invisible to faintly visible, querying all five models at each step. Each model gets its own threshold, calibrated on untouched pairs, since their distances sit on different scales.

Measurements

Main result. The share of same-person pairs a model stops matching, against edit budget, one line per model, alongside an image similarity score (SSIM or LPIPS) at each budget showing how visible the edit had become where protection began working.

Required comparisons. Random noise at the same budget, telling us whether the optimisation contributed anything. JPEG compression and blurring, which any user could apply for free. Untouched pairs, giving each model's starting accuracy. Compression before the model sees the photograph, testing whether ordinary processing removes the edit.

Scope and risks

One editing method, one budget sweep, one trained model. If we fall behind we drop to two test models, keeping one that shares our design and one that differs, which still answers the question.

Main risk. Each model expects images cropped and aligned its own way, and errors here corrupt the comparison while still producing numbers. We confirm every model reaches its published accuracy on untouched pairs before any edit. **Second risk.** Obtaining a training dataset.